**19** Show that $\frac{8+\sqrt{12}}{5+\sqrt{3}}$ can be written in the form $\frac{a+\sqrt{3}}{b}$, where $a$ and $b$ are integers.

$\sqrt{12} = 2\sqrt{3}$

$(8 + 2\sqrt{3})/(5 + \sqrt{3}) \times (5 - \sqrt{3})/(5 - \sqrt{3})$

$= (40 - 8\sqrt{3} + 10\sqrt{3} - 6)/(25 - 3)$

$= (34 + 2\sqrt{3})/22$

$= (17 + \sqrt{3})/11$

**1 mark**
$\sqrt{12} = 2\sqrt{3}$

**1 mark**
multiply by conjugate

**1 mark**
expand numerator/denominator

**1 mark**
$(17+\sqrt{3})/11$

**(Total for Question 19 is 4 marks)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA